Daniyal Ali Hameedi, **Nationality: American and British**, **Date of Birth: 31/10/2001**
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Technical Skills & Languages

- **Programming Languages:** Python, C++, Rust, C#, MATLAB, Lean4, Mathematica, LaTeX.
- **Core Competencies:** Quantitative Modeling, Stochastic Inference, Non-linear Filtering, Monte Carlo Simulations, CUDA GPU Acceleration, Machine Learning, Representation Theory, Conformal Field Theory, Statistical Mechanics.
- **Foreign Languages:** French, Latin, Ancient Greek, Basic Farsi.

Research & Professional Experience

Highforest Capital Ltd London, UK
Quantitative Researcher / Developer 02/2021 – 11/2023, 01/2026 – Present
- Wrote complex tail-risk Monte Carlo simulations for various financing projects using Hidden Markov Model (HMM) approaches. Implemented CUDA GPU acceleration techniques to significantly improve throughput and computational efficiency.

Independent Research London, UK
Independent Researcher 01/2026 – Present
- **Non-linear Filtering / Inference Theory:** Applying the representation theory of Vertex Operator Algebras to classify exactly solvable, infinite-dimensional, Zakai Equation filtering solutions for the cubic-sensor Hidden Markov Model. (Ongoing).
- **Hidden Markov Models**: Formulated the Sector Filter reduction from exponential to polynomial time with accompanying Python implementation uploaded to GitHub.

King's College London London, UK
Research Associate (Part-Time) 06/2025 – 12/2025
- Investigated explicit scaling relations between Yang-Lee and Imaginary Field Ising models using the Functional Renormalization Group to confirm Conformal Field Theory predictions.

Xenophon Systems London, UK
Co-Founder & CEO 12/2022 – 12/2023
- Prototyped a prediction-market mechanism design for eliciting truthful signals about intelligence outcomes in high-leverage low-trust environments without leaking sensitive underlying information.

London Institute for Mathematical Sciences (LIMS) London, UK
Undergraduate Research Assistant (Self-Directed) 05/2023 – 09/2023
- Conducted research on machine learning applications in mathematics and string theory, where we published the open-source `extdivga` machine learning library in Rust. Primarily focused on generating optimal representations for exterior derivative maps.

Education

King's College London London, UK
MSci in Physics and Theoretical Physics 2021 – 2025
- **Grade:** First Class Honours (4.0 GPA equiv.)
- **Thesis:** *Regular and Logarithmic Conformal Invariance in Polymer Gelation (81%)*
- **Academic Achievements:** Super-symmetry and Conformal Field Theory (95%), Quantum Field Theory (89%).

Westminster School London, UK
A-Levels & GCSEs 2015 – 2020
- **A-Levels (2020):** Mathematics (A), Physics (A), Economics (B), Further Mathematics (C).
- **GCSEs (2018):** 7 A*s, 3 As in relevant subjects.

Pre-prints & Publications

- D. A. Hameedi."From Exponential to Polynomial: An Exact Filter for High-Dimensional MSM Models". Pre-print: arXiv:2608.22864, 2026. URL: https://arxiv.org/abs/2608.22864.

Conferences, Competitions & Awards

- **Iliad Intensive Course (10/2026):** 4 week long intensive course focused on mathematically rigorous approaches to AI Safety in the San-Francisco Bay Area.
- **KCL Self-Sourced Research Award (2025):** Awarded for research efforts on the Yang-Lee-Ising Universality Class alongside the Sarkar Research Group.
- **IOP International Physicists' Tournament (2022 – 2023, 2023 – 2024):** Represented the UK in the International Finals at ETH Zurich, leading KCL to its highest-ever placement after placing 1st at UK National Team Selection in 23/24 Tournament. Placed 2nd at UK National Team Selection in 22/23 Tournament.
- **KCL Theoretical Particle Physics & Cosmology Group (2024):** Invited to meetings of Prof. Eugene Lim's research group.
- **String DANGER (07/2023, 07/2024):** 2-Day Conferences on ML applications in Mathematical Research: engaged with the active research community at the intersection of String Theory, Data Science, Number Theory and Geometry.
- **Oxford University ML Summer School (07/2023):** Invited to 5-day bootcamp applying machine learning to string theory and geometry aimed at graduate students and post-docs.
- **Purple and Gold Scholar (2021):** University of Washington, Seattle (awarded highest scholarship level).